

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 07: Tuesday 01 September 2026

(Week 4)

Remark. Q06.4 to Q06.6 are leftover exercises from the Friday class in Week 3.

Remark. Q07.1 to Q07.9 are on Section 13.1-13.3.

Questions on Section 13.7 will be set for this Friday.

Q06.3.5. (Supplementary Exercise).

Consider the function $f(x, y) = x^2 - y$. Sketch the level curves $f(x, y) = k$ for $k = -2, -1, 0, 1, 2$.

Q06.4. Describe the level sets of the following function geometrically:

$$f(x, y, z) = \sqrt{16 - x^2 - y^2 - z^2}.$$

Q06.5. Evaluate the following limits:

(i) $\lim_{(x,y) \rightarrow (0,\pi/4)} \sec x \tan(x + y).$

(ii) $\lim_{(x,y) \rightarrow (2,0)} \frac{\sqrt{2x - y} - 2}{2x - y - 4}.$

[Remark: For (ii), you may find the identity $a^2 - b^2 = (a + b)(a - b)$ useful.]

Q06.6. Does the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{y}{x^2}$ exist? Justify your answer.

Q07.1. Consider the function

$$f(x, y) = \begin{cases} \sqrt{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

(i) Find $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$.

(ii) Is the function $f(x, y)$ continuous at $(0, 0)$? Justify your answer.

Q07.2. (a) Find the first partial derivatives of each of the following functions:

(i) $f(x, y) = 4x^3 - 3x^2y^2 + 2x$.

(ii) $f(x, y) = \cos(3x^2y + 5y)$.

(iii) $f(x, y) = y \ln(4x + 3y)$.

(iv) $f(x, y, z) = y^2z + x^3yz^3$.

(b) (i) For the function in (a) (iii), find also the values of the first partial derivatives at the point **(2, 1)**.

(ii) For the function in (a) (iv), find also the values values of the first partial derivatives at the point **(1, 2, 4)**.

Q07.3. Using implicit differentiation, find $\frac{\partial \mathbf{z}}{\partial \mathbf{x}}$ and $\frac{\partial \mathbf{z}}{\partial \mathbf{y}}$ if

$$\mathbf{x}^3 + 2\mathbf{y}^3 + 6\mathbf{x}\mathbf{z}^3 - 2 = 59\mathbf{z}.$$

[Remark: Here we think of $\mathbf{z}$ as a function of the two independent variables $\mathbf{x}$, $\mathbf{y}$ defined implicitly by the above equation. In other words, we think of an implicitly defined function $\mathbf{z}(\mathbf{x}, \mathbf{y})$.]

Q07.4. Consider the function $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \sin(\mathbf{x} + \mathbf{y}^2)$.

(i) Evaluate $\mathbf{f}_\mathbf{x}$, $\mathbf{f}_\mathbf{y}$, $\mathbf{f}_{\mathbf{x}\mathbf{x}}$, $\mathbf{f}_{\mathbf{x}\mathbf{y}}$, $\mathbf{f}_{\mathbf{y}\mathbf{x}}$, $\mathbf{f}_{\mathbf{y}\mathbf{y}}$.

(ii) What do you observe about $\mathbf{f}_{\mathbf{x}\mathbf{y}}$ and $\mathbf{f}_{\mathbf{y}\mathbf{x}}$?

Q07.5. Verify the identity $f_{xy} = f_{yx}$ for the function

$$f(x, y) = x^3y^4 + e^{x+2y}.$$

Q07.6. Suppose $f(x, y)$ is a function such that

$$f_x = 3x^2 + Axy, \quad f_y = 3x^2 + 6y^2,$$

where A is a constant.

(i) Using Clairaut's Theorem (i.e. the identity $f_{xy} = f_{yx}$), determine the value of A .

(ii) Using your result in (i), find such a function $f(x, y)$ by inspection or any method using your experience in one variable calculus.

[Remark. Such a function is not unique.]

Q07.7. Let $f(\mathbf{x}, \mathbf{y})$ be a function, and $(\mathbf{x}_o, \mathbf{y}_o)$ be a point in the domain of f . Consider the linear polynomial (in two variables) given by

$$g(\mathbf{x}, \mathbf{y}) = \mathbf{a} + \mathbf{b}(\mathbf{x} - \mathbf{x}_o) + \mathbf{c}(\mathbf{y} - \mathbf{y}_o)$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are constants.

(i) Determine the values of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ (in terms of data on f) if

$$g(\mathbf{x}_o, \mathbf{y}_o) = f(\mathbf{x}_o, \mathbf{y}_o), \quad g_x(\mathbf{x}_o, \mathbf{y}_o) = f_x(\mathbf{x}_o, \mathbf{y}_o)$$

$$\text{and} \quad g_y(\mathbf{x}_o, \mathbf{y}_o) = f_y(\mathbf{x}_o, \mathbf{y}_o).$$

(ii) Write down equation for the graph of $g(\mathbf{x}, \mathbf{y})$.

[Remark: Note that the equation in (ii) is simply an equation of the tangent plane to the graph of $f(\mathbf{x}, \mathbf{y})$ at the point $(\mathbf{x}_o, \mathbf{y}_o)$ as given in Prof Auroux's notes, Lecture 9.]

Q07.8. Consider the function $f(x, y) = x^2y + 2xy$. Find an equation of the tangent plane to the graph of $f(x, y)$ at the point $(1, 2)$.

[Remark: Recall the result in the previous question.]

Q07.9. Consider the surface in $\mathbb{R}^3$ defined by the equation $yz = x^2$. Find an equation of the tangent plane to the surface at the point $(4, 2, 8)$.

[Suggestion: First rewrite the surface as the graph of a function of two variables.]